Programming the serial port

Modifying the serial port for scientific test devices

This article explains how to modify the serial port's data word length, stop bits, and parity setting. Changing the Osborne 1's default specification of 8-bit data word, 1 stop bit and no parity is useful when interfacing to certain scientific test devices that have limitations in their interfacing options. These steps are simple and will take only a few minutes to perform.

The change permanently modifies the operating system (BIOS) with a new default value for the serial port. When you boot with the modified diskette, the modification is loaded.

To begin, you will need a formatted diskette in the B drive, and the following CP/M utilities on an otherwise blank diskette, with system tracks, in the A drive:

1. DDT.COM
2. SYSGEN.COM
3. MOVCPM.COM

Note: This modification is for the "1.3"BIOS. If you have a late model Osborne, or have the double density option installed, you may have a "1.4" BIOS. See Figure 1 for details.

If you have a 1.3 BIOS use **MOVCPM 6Ø** and **CMP6Ø.COM** as indicated below; else, substitute **MOVCPM 59 *** and **CPM59.COM** in the case of a 1.4 BIOS.

```
A > MOVCPM 6Ø *

CONSTRUCTING 6ØK CP/M VERS 2.2
READY FOR "SYSGEN" OR
"SAVE 39 CPM6Ø.COM"
A>SAVE 39 CPM6Ø.COM
A>DDT CPM6Ø.COM
DDT VERS 2.2
NEXT PC
28ØØ Ø1ØØ
-S1FE9
1FE9 55   xx   (xx is the code you are inserting.
1FEA 8Ø   .    Refer to Figure 1 for a table
- ^C           of 685Ø control codes.)
A>SYSGEN

        Operating System Generation Program
        OSBORNE COMPUTER SYSTEM ONE

SOURCE drive (A or B) <cr>  (reads from memory)

DESTINATION (A,B or RETURN to exit) B
Put DESTINATION diskette in B, then press RETURN
<cr>

DESTINATION (A,B OR RETURN TO EXIT) <cr>
```

The diskette in the B drive is now modified with the serial control code you specified.

System Technical Support Group, Osborne Computer Corporation

WHICH BIOS DO YOU HAVE?

If you don't know the answer to that question, do the following in order to find out which BIOS you are working with.

Boot up your utility system and then:

A>**DDT**
DDT VERS 2.2
-**DØØØØ**

ØØØØ C3 Ø3 E5 (etc.) .. ..
ØØØ8 (etc.)

ØØØØ C3 Ø3 E1 (etc.) .. ..
ØØØ8 (etc.)

You should see something like one of the above. If in the third location following the ØØØØ there is an E5, you have a 1.3 BIOS; if there is an E1, then you have a 1.4 BIOS.

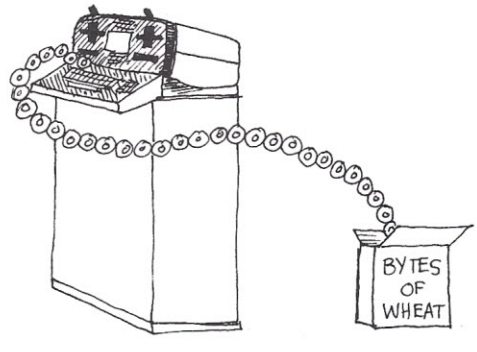
Figure 1

TABLE OF 6850 CONTROL CODES

data bit	parity	stop bit	baud rate	xx Hex
7	even	2	1200	41
7	odd	2	1200	45
7	even	1	1200	49
7	odd	1	1200	4D
8	none	2	1200	51
→8	none	1	1200	55
8	even	1	1200	59
8	odd	1	1200	5D
data bit	parity	stop bit	baud rate	xx Hex
7	even	2	300	42
7	odd	2	300	46
7	even	1	300	4A
7	odd	1	300	4E
8	none	2	300	52
→8	none	1	300	56
8	even	1	300	5A
8	odd	1	300	5E

The arrows point to the standard default settings. For further information on other codes, consult the Motorola 6850 specification sheet supplied in the Osborne 1 Technical Manual.

Figure 2



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